

$$\frac{1}{s^2 R_c} \rightarrow \frac{V_{dc}}{R_c} \times$$

$$\tau \frac{dy}{dt} + y = k_p f(t)$$

$$L(y) = Y(s) \longrightarrow \text{Output}$$

$$L\left(\frac{dy}{dt}\right) = sY(s) - y(0) = sY(s)$$

$$\rightarrow \tau (sY(s)) + Y(s) = k_p L(f(t))$$

$$\Rightarrow \frac{(\tau s + 1) Y(s)}{k_p} = L(f(t)) \longrightarrow \text{Input}$$

$$H(s) = \frac{\text{Output}}{\text{Input}} = \frac{Y(s) k_p}{(\tau s + 1) Y(s)} = \frac{k_p}{(\tau s + 1)}$$